



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. CSE 'A'

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Lists, Tuples, Dictionaries

- **Course Outcome:** CO4

- **Topic Learning Outcome:** TLO11

- **Activity Chosen:** Crossword Puzzle

- **Justification:**

Different types of data structures can be used to store data of different kinds. The students have to identify the correct data structures to be used in various applications and the methods used to operate with them. Crossword puzzle helps them to recollect the different methods associated with the particular data structures.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

- A crossword puzzle is a fun word game that encourages students to remember the idea and try to figure out the solution on their own. It encourages students' curiosity instead of giving them the typical MCQ exams.
- The activity was conducted in a classroom for a duration of 10 minutes. The students were given 12 questions as a crossword puzzle in the topic of list, tuple and dictionary. They have to think and write the answers in the given puzzle.
- The activity was conducted during the last 20 minutes of the class after that the answers were corrected. Sample crossword puzzles solved by the students were shown in figure 1.
- At the end of class, the correct answers to the incorrect questions were discussed to ensure that everyone understood the correct responses to all the questions.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	3	2	2	2	2	1	1	1	3	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO3
	3	2	2	1	1	3
Justification for correlation	The students apply the	The students analyze	The students find	The students demonstrate	The students communicate the solution	The students choose and apply the

	knowledge of data structures such as list, tuple and dictionary	the given questions and identify solutions	solutions for the given question.	problem-solving skills as a team.	after solving the puzzle	correct data structures for solving the problems in AI
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• **Images / Screenshot of the practice:**

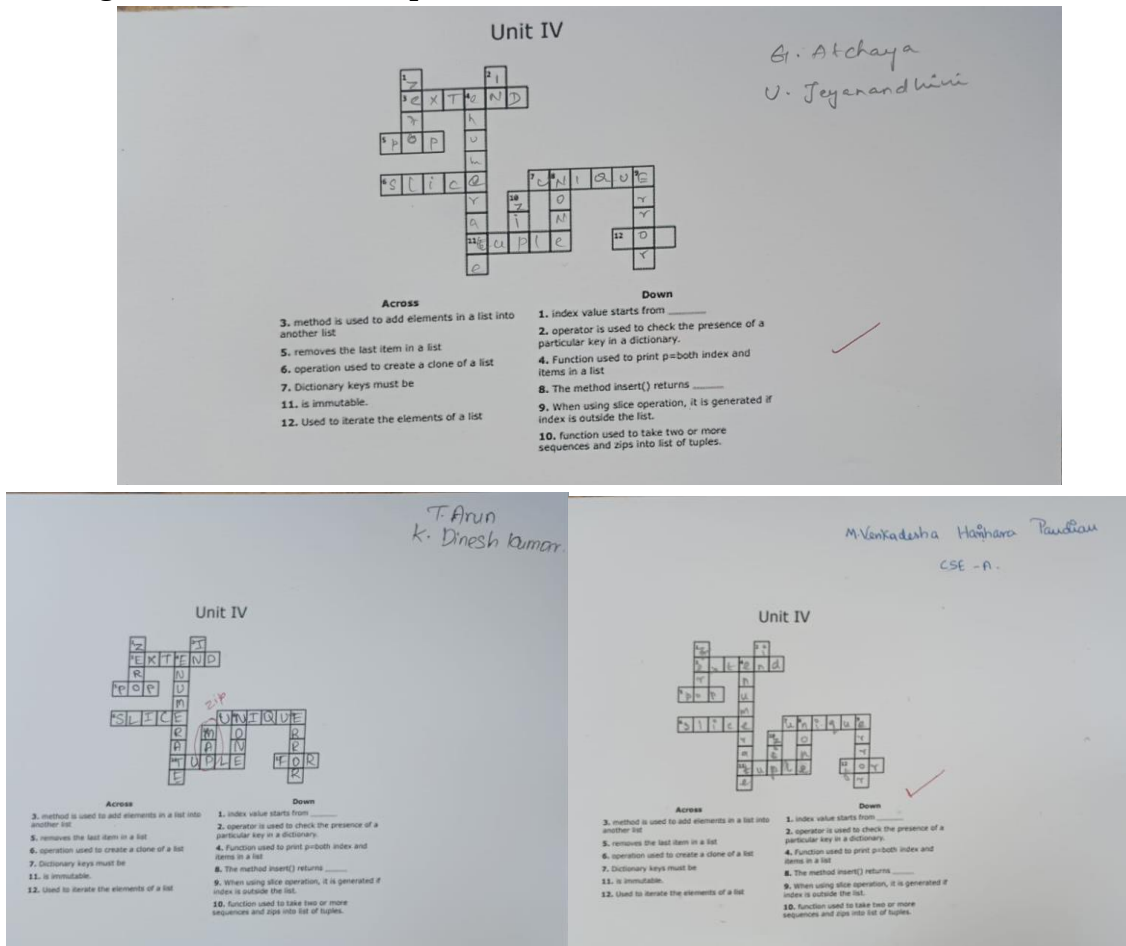


Fig 1: Sample Answers

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Some of the students find very easy to identify the methods used with list, tuple and dictionary.
- Some of them take more time to complete the puzzle need more clarification on the some of the methods.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to think on their own and identify the how the method works and its return value and based on that how to choose which data structure to solve the problems. The students level of understanding on this topic gets increased.

❖ ***Challenges faced in implementation:***

- The students those who didn't clearly understand the basics and the various methods of list and dictionary were struggled to find the answer for the questions.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/48.GE8151_Crossword.pdf

Signature of Faculty Member

HOD